

- User accounts
- Inventory
- E-commerce transactions
- Experience customization
- User-created content
- Site data

If the phrase *site data* seems vague to you, that's because it is. Many websites store and display data that is specific to that site. For example, a movie site has data for movies, and a book site has data for books. For Nicky's RiffMates site you need

- Musicians
- Bands
- Classifieds
- Venues
- Venue lineups

As RiffMates grows, so will this list. There are a variety of approaches for storing this data, but by far, the most common is to use a relational database. You'll recall that a relational database stores content in a series of tables similar to how a spreadsheet works. The columns in a table have a data type, typically storing strings or numbers. To represent a band, your columns might include the band name and the date they were formed. Each row in the table corresponds to a single band. To be able to reference a band, the table includes an identity column, typically an auto-incrementing number provided by the database. This identity is called the table's *primary key*.

Columns can also indicate a relationship between data items, hence the term *relational* in *relational database*. To represent a song, your columns include the name of the song and, like the bands, a unique identifier. To link a band and a song, either the band or song table includes a column that references the other's unique identifier. The unique identifiers are often *primary keys*: special numbers the database creates for you. You can see an example of object relationships in figure 4.1.

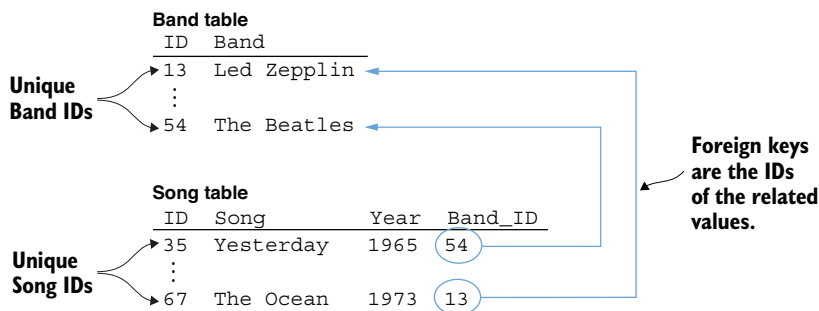


Figure 4.1 Inter-table relationships between bands and songs